

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

Correct Answer: 16

Rationale

The correct answer is **16**. An equation of a circle in the xy -plane can be written as $(x - t)^2 + (y - u)^2 = r^2$, where the center of the circle is (t, u) , the radius of the circle is r , and where t , u , and r are constants. It's given that the equation of circle A is $(x + 5)^2 + (y - 5)^2 = 4$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 2^2$. Therefore, the center of circle A is $(-5, 5)$ and the radius of circle A is **2**. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $(-5, 5)$ and the radius of circle B is **2(2)**, or **4**. Substituting -5 for t , 5 for u , and **4** for r into the equation $(x - t)^2 + (y - u)^2 = r^2$ yields $(x + 5)^2 + (y - 5)^2 = 4^2$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 16$. It follows that the equation of circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = 16$. Therefore, the value of k is **16**.